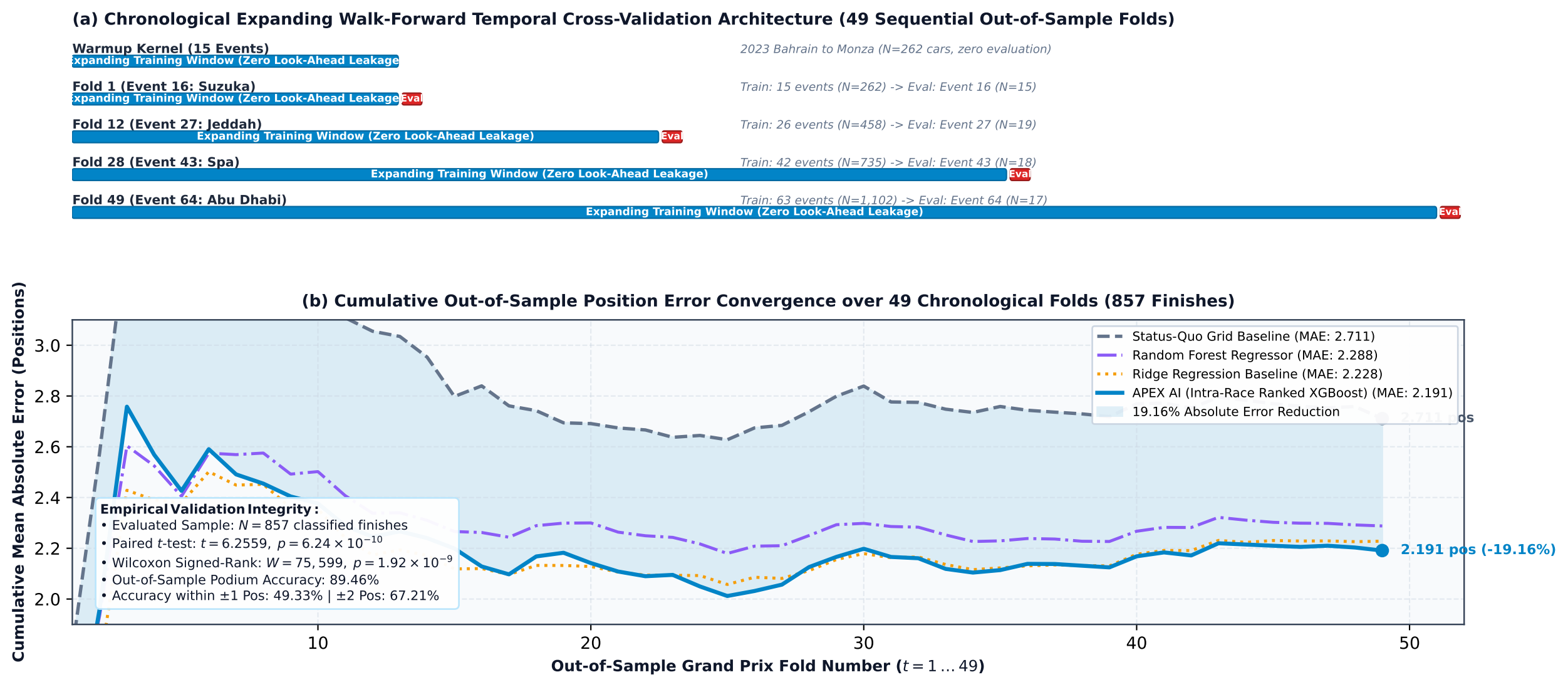

(a) Chronological Expanding Walk-Forward Temporal Cross-Validation Architecture (49 Sequential Out-of-Sample Folds)
Warmup Kernel (15 Events)
Expanding Training Window (Zero Look-Ahead Leakage)
2023 Bahrain to Monza (N=262 cars, zero evaluation)
Fold 1 (Event 16: Suzuka)
Expanding Training Window (Zero Look-Ahead Leakage)
Eval
Train: 15 events (N=262) -> Eval: Event 16 (N=15)
Fold 12 (Event 27: Jeddah)
Expanding Training Window (Zero Look-Ahead Leakage)
Eval
Train: 26 events (N=458) -> Eval: Event 27 (N=19)
Fold 28 (Event 43: Spa)
Expanding Training Window (Zero Look-Ahead Leakage)
Eval
Train: 42 events (N=735) -> Eval: Event 43 (N=18)
Fold 49 (Event 64: Abu Dhabi)
Expanding Training Window (Zero Look-Ahead Leakage)
Eval
Train: 63 events (N=1,102) -> Eval: Event 64 (N=17)
(b) Cumulative Out-of-Sample Position Error Convergence over 49 Chronological Folds (857 Finishes)
Cumulative Mean Absolute Error (Positions)
3.0
2.8
2.6
2.4
2.2
2.0
10
20
30
40
50
Out-of-Sample Grand Prix Fold Number ($t = 1 \dots 49$)
Status-Quo Grid Baseline (MAE: 2.711)
Random Forest Regressor (MAE: 2.288)
Ridge Regression Baseline (MAE: 2.228)
APEX AI (Intra-Race Ranked XGBoost) (MAE: 2.191)
19.16% Absolute Error Reduction
2.711 pos
2.191 pos (-19.16%)
Empirical Validation Integrity :
• Evaluated Sample: $N = 857$ classified finishes
• Paired t-test: $t = 6.2559$, $p = 6.24 \times 10^{-10}$
• Wilcoxon Signed-Rank: $W = 75,599$, $p = 1.92 \times 10^{-9}$
• Out-of-Sample Podium Accuracy: 89.46%
• Accuracy within ±1 Pos: 49.33% | ±2 Pos: 67.21%